PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Nurse-led vs. usual-care on cardiovascular risk factors for patients with type 2 diabetes: a systematic review and meta-analysis
AUTHORS	Zhang, Jiayu; Zheng, Xutong; Ma, Danyan; Liu, Changqin; Ding, Yulan

VERSION 1 – REVIEW

REVIEWER	Chang, Yue-Fang University of Pittsburgh Department of Neurological Surgery
REVIEW RETURNED	18-Nov-2021

GENERAL COMMENTS	This manuscript by Zhang J. et al. employed meta-analysis to evaluate the effectiveness of nurse-led care (vs. usual care) on cardiovascular risk factors among type 2 diabetes patients. Thirteen articles with a total of 3475 patients were included in the meta-analysis. 1. A major concern of this manuscript is that, to study the effect of nurse-led intervention, rather than assessing the change scores from baseline to follow up, the authors only compared the outcome measures at follow up between the nurse-led care and usual care groups. Even though all the studies included in the meta-analysis were randomized trials, baseline measurements still need to be considered in the comparison. One example of this is that in the study by Al Lenjawi B. et al. (ref 32), the intervention group had much higher LDL at baseline, but by 12 months of follow up, the LDL were similar in the 2 groups (Figure 2C). If one only looked at 12-month data, it appeared that the intervention and control groups had similar effects; but if comparing the change score in LDL, the intervention group had much greater reduction. 2. The follow up time of the 13 studies ranged from 6 months to 30 months. Please comment on pooling the results together with such a wide range of follow up time. 3. The numbers presented in the Searching Process section were confusing and did not match the numbers listed on Figure 1. 4. For the SBP results (Page 14 lines 38-43): It mentioned that “5 studies (Tang PC, et al,2013; Jansink R, et al, 2013; Morgan MA, et al,2013; Azami G, et al,2018; and Jayasuriya R, et al. 2015) reported that there was a slight improvement in SBP in the follow-up months.” I assumed the authors meant that 5 of the studies showed lower SBP at follow up in the experimental group when compared to the control group. However, according to Figure 2D, the studies by Jansink and by Morgan had experimental group with higher SBP. Please clarify. 5. The authors concluded that “the overall effect size was 0.58 of a standard deviation improvement in HbA1c compared to usual

	care...” (page 16 line 43) Please explain where this ‘0.58 of a standard deviation improvement’ came from. 6. Please include reference for the statement “I2>50% was considered high heterogeneity …”. 7. It is not clear whether random effect model or fixed effect model was used for the BMI analysis which had an I2 of 44%. Figure 2B forest plot showed it was based on random effect model. But according to the method section, a fixed effect model was used if I2 < 50% 8. Please make sure the information regarding the studies by Odnoletkova and by Al Lenjawi was correctly listed. 9. Pooled rate ratios and funnel plot were mentioned in the Data synthesis and analysis section. But no results were listed for rate ratio and no funnel plots were presented to assess the publication bias. 10. Negative signs were omitted for the confidence intervals on page 13 line 54. 11. What is WMD? 12. Figure 2C: please include the unit of LDL.

REVIEWER	Homko, Carol Temple University
REVIEW RETURNED	06-Dec-2021

GENERAL COMMENTS	This systemic review and metanalysis examined the effectiveness of nurse-led interventions as compared to usual care on cardiovascular risk factors among patients with type 2 diabetes. Specific Comments: 1. It is not clear to this reviewer that the nurse-led interventions/programs analyzed in this review are in fact comparable. From both a clinical and intervention standpoint there is a significant difference between diabetes self-management education, SMS, telephone counseling and NLTM. Not to mention the and intervention fact that some interventions were face to face while others were remote. 2. Another major concern is the primary focus on A1C to the almost exclusion of other CVD risk factors. Neither their abstract conclusion nor their summary of strengths and limitations address risk factors other than A1c. I would have anticipated a more in-depth discussion as to why other CVD risk factors were not impacted by these interventions. 3. The text of the manuscript states that the nurses represented in these studies were nurse managers, clinic nurses and community-based nurses. Did these nurses have specialized training in diabetes education and management? Were these advanced nurse practitioners who were actually managing the patient’s diabetes as well as their hypertension, hypercholesterolemia etc.? It does not appear to be the case for most of these studies. In fact, in their text the authors state “nurses in most countries are unable to prescribe or titrate treatments.” Although not impossible it is difficult to treat hypertension and hyperlipidemia by life-style alone. 4. The authors only report change in the outcomes of interest. Clearly if patients entered the study with an A1C within target or close to target (the same applies for any of the outcomes), it would be difficult to show a significant change. It is important to know if patients were referred to these programs for suboptimal control. 5. Lastly, I would recommend the manuscript be edited by someone fluent in English.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Comments to the Author:
C: This manuscript by Zhang J. et al. employed meta-analysis to evaluate the effectiveness of nurse-led care (vs. usual care) on cardiovascular risk factors among type 2 diabetes patients. Thirteen articles with a total of 3475 patients were included in the meta-analysis.
R: We appreciate your comments very much and revised the manuscript carefully.

• Major comments:
C1: A major concern of this manuscript is that, to study the effect of nurse-led intervention, rather than assessing the change scores from baseline to follow up, the authors only compared the outcome measures at follow up between the nurse-led care and usual care groups. Even though all the studies included in the meta-analysis were randomized trials, baseline measurements still need to be considered in the comparison. One example of this is that in the study by Al Lenjawi B. et al. (ref 32), the intervention group had much higher LDL at baseline, but by 12 months of follow up, the LDL were similar in the 2 groups (Figure 2C). If one only looked at 12-month data, it appeared that the intervention and control groups had similar effects; but if comparing the change score in LDL, the intervention group had much greater reduction.
R1: We appreciate your comments very much. According to your nice suggestions, we have re-analyzed the data based on the change of cardiovascular risk factors from
baseline measurements. Except for HbA1c, we also found the change scores from baseline to follow up in both BMI and SBP have significant differences between two groups. But reduction of LDL-c still has no significnce. We have revised the abstract, results section and discussion
section accordingly in the updated manuscript. The updated the results were shown in Figure 4.

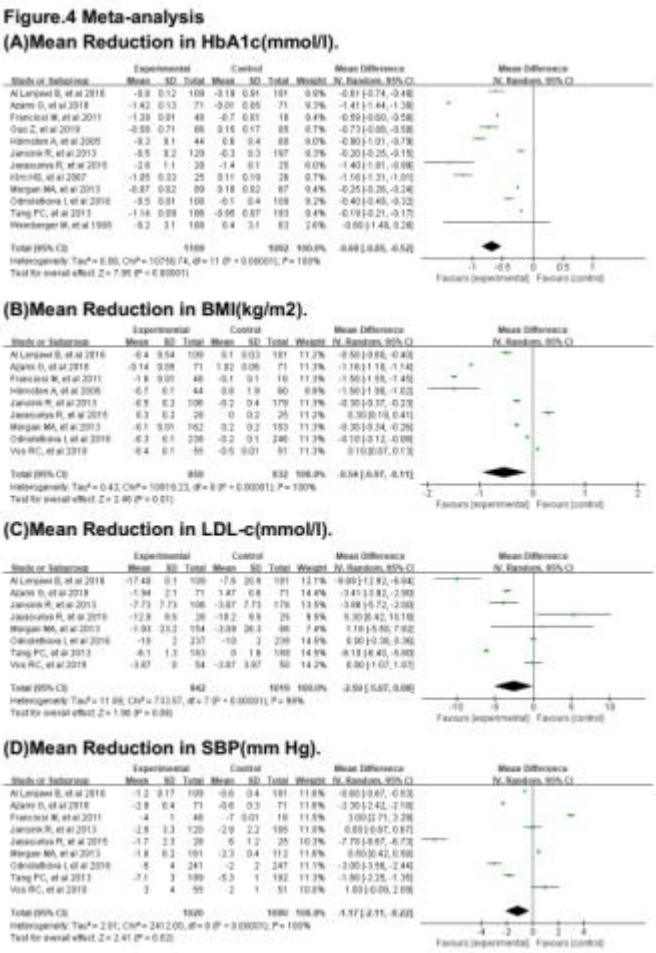


Figure 4. Comparison of the change scores of cardiovascular risk factors from baseline to follow up

C2: The follow up time of the 13 studies ranged from 6 months to 30 months. Please comment on pooling the results together with such a wide range of follow up time.
R2: We appreciate the comments. Most of the articles (12/13) included for meta-analysis followed up for 6-18 months, only in one study (vos R C, et al; Diabet Med. 2019;36(7):827-835.) the follow up time was up to 30 months. We have added this information as one limitation.

Page 21
Another limitation might be that the follow up time of the 13 studies ranged from 6 months to 30 months.

C3: The numbers presented in the Searching Process section were confusing and did not match the numbers listed on Figure 1.
R3: We appreciate the comment very much and have updated the searching process section and the numbers listed on Figure 1.

Page 11
A total of 2,455 articles were retrieved. After the removal of duplicates, 1,589 articles were retained for abstract and title screening. Initial screening excluded 1,403 articles; only 186 articles remained for full text screening. Further, 173 articles were excluded due to various reasons such as non-nurse-led (n=59), non-eligible population types (n =26), less than 6 months of follow-up (n=18), conference abstract (n=15), non-English articles (n=10), not RCTs (n=13), duplicate reporting (n=9), unrelated to the pre-decided outcomes (n=2)

C4: For the SBP results (Page 14 lines 38-43): It mentioned that “5 studies (Tang PC, et al,2013; Jansink R, et al, 2013; Morgan MA, et al,2013; Azami G, et al,2018; and Jayasuriya R, et al. 2015) reported that there was a slight improvement in SBP in the follow-up months.” I assumed the authors meant that 5 of the studies showed lower SBP at follow up in the experimental group when compared to the control group. However, according to Figure 2D, the studies by Jansink and by Morgan had experimental group with higher SBP. Please clarify.
R4: We appreciate this comment and have corrected literatures.

Page 16
Among these studies, 3 studies (Tang PC, et al,2013; Azami G, et al,2018; and Jayasuriya R, et al,2015)[14, 18, 32] reported that there was a slight improvement form baseline in SBP in the follows-up months in the experimental group when compared to the control group. Other studies (Franciosi M, et al,2011; Odnoletkova I, et al,2016; Morgan MA, et al,2013; Jansink R, et al, 2013; Al Lenjawi B, et al,2016; and Vos RC, et al,2019)[19, 20, 31, 33-35] found that there was no significant difference in SBP between the groups ($p>0.05$).

C5: The authors concluded that “the overall effect size was 0.58 of a standard deviation improvement in HbA1c compared to usual care...” (page 16 line 43) Please explain where this ‘0.58 of a standard deviation improvement’ came from.
R5: We appreciate this comment. The standard deviation improvement was come from the change scores of cardiovascular risk factors from baseline to follow up. We have re-analyzed the data based on the change scores from baseline to follow up and the updated standard deviation improvement in HbA1c was 0.68.

Page 17
The overall effect size should be 0.68 of a standard deviation improvement in HbA1c compared to usual care (Fig.4(A)),

C6: Please include reference for the statement “I^2>50% was considered high heterogeneity …”.
R6：We appreciate this comment and have added the information based on the reference 26 (Stat Med. 2002;21(11):1539-1558.) and reference 27 (BMJ. 2003;327(7414):557-560.).

Page 10
I^2>50% was considered high heterogeneity (random-effects model was used), and values lower than 50% to indicate low heterogeneity (fixed effects model was used) [26, 27].

C7: It is not clear whether random effect model or fixed effect model was used for the BMI analysis which had an I² of 44%. Figure 2B forest plot showed it was based on random effect model. But according to the method section, a fixed effect model was used if I² < 50%
R7: We appreciate this comment. We have updated the results based on the change scores from baseline to follow up. In the revised manuscript and we still used random effect model based on the updated I^2 of 100%.

C8: Please make sure the information regarding the studies by Odnoletkova and by Al Lenjawi was correctly listed.
R8: We appreciate this comment and have corrected the legend of Table 1.

C9: Pooled rate ratios and funnel plot were mentioned in the Data synthesis and analysis section. But no results were listed for rate ratio and no funnel plots were presented to assess the publication bias.

R9: We appreciate this comment and have added the funnel plots in Supplementary Materials 3. Since the main outcome measures were continuous variables, we did not calculate the rate ratio in the analysis and we deleted this information in the method section in the revised manuscript.

Page 10
We have calculated weighted mean differences (WMD) with 95% confidence intervals (95% CI) and pooled values for rates presented as rate ratios.

C10: Negative signs were omitted for the confidence intervals on page 13 line 54.
R10: We appreciate this comment. Since we have re-analyzed and found different results, sensitivity analysis by individual trial exclusion showed that one trials (Weinberger M; J Gen Intern Med. 1995;10(2):59-66.) have the greatest effect on the result of meta-analysis. After excluding the trial, the heterogeneity for HbA1c were no reduced. Thus, we have deleted the expressions.

Page 14
Sensitivity analysis by individual trial exclusion showed that two trials (Azami G, et al,2018, Jansink R, et al, 2009)[13, 26] have the greatest effect on the mean difference and heterogeneity. After excluding the above two trials the size effect and heterogeneity for HbA1c were significantly reduced (WMD=-0.54; 95% CI: 0.64, 0.44, $p<0.001$; $I^2=54\%$, $p<0.001$).

C11: What is WMD?
R11: We appreciate this comment. WMD means weighted mean difference, which was used when all studies in the Meta-analysis have the same continuous outcome variable (such as weight) and measurement unit.

Page 10
We have calculated weighted mean differences (WMD) with 95% confidence intervals (95% CI)

C12: Figure 2C: please include the unit of LDL.
R12: We appreciate this comment and have added the unit of LDL (mmol/l) in Figure.4(C).

Reviewer: 2
Comments to the Author:
C: This systemic review and metanalysis examined the effectiveness of nurse-led interventions as compared to usual care on cardiovascular risk factors among patients with type 2 diabetes.
R: We appreciate your comments very much and revised the manuscript carefully.

Specific Comments:
C1: It is not clear to this reviewer that the nurse-led interventions/programs analyzed in this review are in fact comparable. From both a clinical and intervention standpoint there is a significant difference between diabetes self-management education, SMS, telephone counseling and NLTM. Not to mention the and intervention fact that some interventions were face to face while others were remote.
R1: We appreciate this comment very much. Nurse-led care is attractive as it has been associated with better adherence to protocols, improved prescribing in concordance with guidelines and more regular follow-up. Whether diabetes self-management education (J Diabetes Res. 2018; 2018:4930157.), SMS (J Clin Nurs. 2007;16(6):1082-1087.), telephone counseling (Diabet Med. 2016;33(6):777-785.) and NLTM interventions (Patient Prefer Adherence. 2019; 13:1353-1362.), their common features are that they all emphasized the leading role of nurses as well as helping patients adhere to treatment plans from different perspectives. Nurses have a key role in patients with

T2DM in delivering patient education, explaining treatment strategies, supporting patients in making lifestyle changes, and maintaining self-management.

In subgroup analysis, nurse-led clinic also had positive impacts on controlling HbA1c level, no matter in clinical setting (WMD = -0.83mmol/l; 95% CI -1.28, -0.37, $p < 0.001$), primary care setting (WMD = -0.62mmol/l; 95% CI -1.11, -0.14, $p = 0.01$) or remote (WMD = -0.58mmol/l; 95% CI -0.94, -0.21, $p = 0.002$).

C2: Another major concern is the primary focus on A1C to the almost exclusion of other CVD risk factors. Neither their abstract conclusion nor their summary of strengths and limitations address risk factors other than A1c. I would have anticipated a more in-depth discussion as to why other CVD risk factors were not impacted by these interventions.
R2: We appreciate your comments very much. We have re-analyzed the data based on the change of cardiovascular risk factors from baseline measurements. Except for HbA1c, we also found the change scores from baseline to follow up in both BMI and SBP have significant differences between two groups. But reduction of LDL-c still has no significance. We have revised the abstract, results section and discussion section accordingly in the updated manuscript. The updated the results were shown in Figure 4.

C3: The text of the manuscript states that the nurses represented in these studies were nurse managers, clinic nurses and community-based nurses. Did these nurses have specialized training in diabetes education and management? Were these advanced nurse practitioners who were actually managing the patient's diabetes as well as their hypertension, hypercholesterolemia etc.? It does not appear to be the case for most of these studies. In fact, in their text the authors state "nurses in most countries are unable to prescribe or titrate treatments." Although not impossible it is difficult to treat hypertension and hyperlipidemia by life-style alone.
R3: We appreciate this comment very much. We strictly follow the inclusion and exclusion criteria, all the nurses included in the analysis articles have been specially trained to provide nurse-led care. For example, in Kim's research (J Clin Nurs. 2007;16(6):1082-1087.), the researcher was a professor in a nursing college and a diabetic educator. The researcher was familiar with the current diabetes guidelines and were specialized to treat T2DM patients. Another example, in Morgan's study (BMJ Open. 2013;3(1): e002171.), theusing practice nurses (PNs) as the identified case manager to undertake the care of the patients. It also includes the use of evidence-based guidelines, systematic screening and monitoring of risk factors, timetabled recall visits, information support for the clinician, enhanced patient self-management, a mean of effective communication between all members of the care team and audit information for the practice. Lastly, nurse in most countries is unable to prescribe but they can help patients adhere to treatment plans and provide information to help patients implement useful techniques. As the saying goes, "three points depend on treatment, and seven points depend on nursing". Diabetes nursing is extremely important to diabetic patients, which is attractive as it has been associated with better adherence to protocols, improved prescribing in concordance with guidelines and more regular follow-up. Therefore, we conducted a systematic review and meta-analysis to evaluate the impact of a nurse-led interventions program versus usual-care on patients with T2DM and provide trustworthy practice guidelines on newly released evidence in the current study.

C4: The authors only report change in the outcomes of interest. Clearly if patients entered the study with an A1C within target or close to target (the same applies for any of the outcomes), it would be difficult to show a significant change. It is important to know if patients were referred to these programs for suboptimal control.
R4: We appreciate your comments very much. We have re-analyzed the data based on the change of cardiovascular risk factors from baseline measurements. Except for HbA1c, we also found the change scores from baseline to follow up in both BMI and SBP have significant differences between two

groups. But reduction of LDL-c still has no significance. We have revised the abstract, results section and discussion section accordingly in the updated manuscript. The updated the results were shown in Figure 4.

C5: Lastly, I would recommend the manuscript be edited by someone fluent in English.
R5: We appreciate the comment very much and have carefully corrected some grammatical errors.